

Tech & Teach

Digital Chip Design Program

RTL Design

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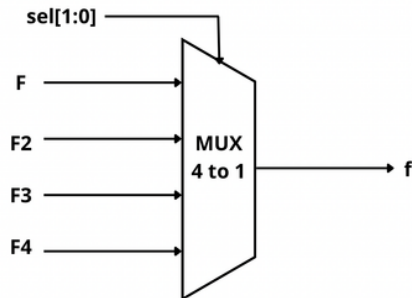
Seq & Comb Circuits Datapath – Part 2

Version 1.0

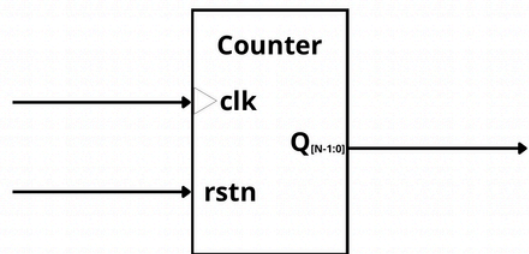
Task #11.1

List the needed components and what is the purpose of each of them

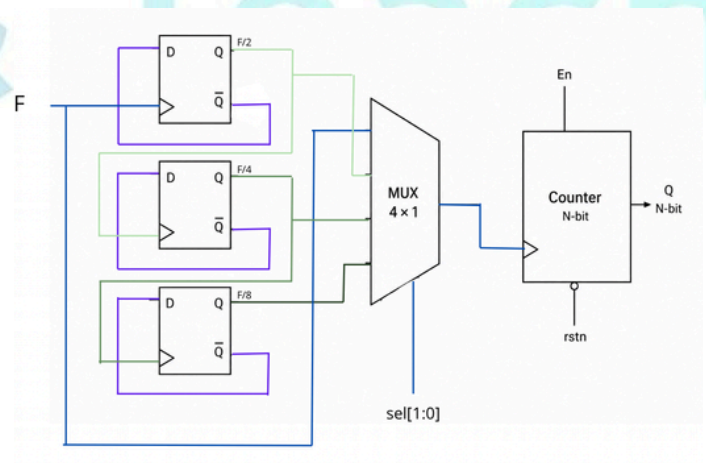
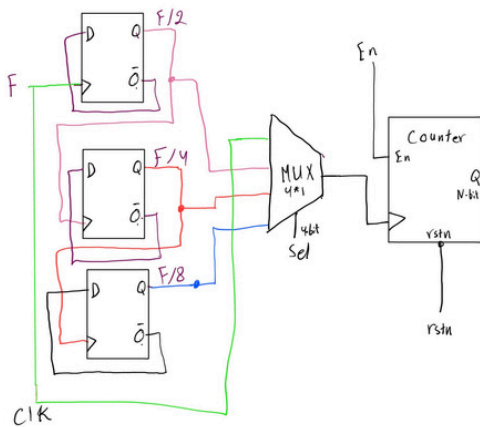
(MUX): Selects one clock from the available frequencies based on sel.



Counter: Counts using the selected clock and outputs an N-bit count (Q).



Clock Divider: Generates slower clock frequencies (F/2, F/4, F/8).



Task #11.1

```
module clockDivider (F, rstn, F2, F4, F8);
input F, rstn;
output reg F2, F4, F8;
always @(posedge F, negedge rstn)
begin
    if(!rstn)
        F2 <= 0;
    else
        F2 <= ~F2;
end
always @(posedge F2, negedge rstn)
begin
    if(!rstn)
        F4 <= 0;
    else
        F4 <= ~F4;
end
always @(posedge F4, negedge rstn)
begin
    if(!rstn)
        F8 <= 0;
    else
        F8 <= ~F8;
end
endmodule
```

```
module Counter #(parameter N = 4)(
    input logic clk,
    input logic rstn,
    output logic [N-1:0] Q
);
always @(posedge clk, negedge rstn)
begin
    if(!rstn)
        Q <= 0;
    else
        Q <= Q + 1;
end
endmodule
```

```
module mux4to1 (
    input logic F,
    input logic F2,
    input logic F4,
    input logic F8,
    input logic [1:0] sel,
    output logic clkOut
);
always_comb begin
    if (sel == 2'b00)
        clkOut = F;
    else if (sel == 2'b01)
        clkOut = F2;
    else if (sel == 2'b10)
        clkOut = F4;
    else if (sel == 2'b11)
        clkOut = F8;
    else
        clkOut = 1'b0;
end
endmodule
```

Task #11.1

```
module top #(parameter N = 4)
```

```
(
```

```
    input logic clk,  
    input logic rstn,  
    input logic [1:0] sel,  
    output logic [N-1:0] Q
```

```
);
```

```
    logic F2, F4, F8;  
    logic selected_clk;
```

```
    clockDivider clkDiv(
```

```
        .F(clk),  
        .rstn(rstn),  
        .F2(F2),  
        .F4(F4),  
        .F8(F8)
```

```
);
```

```
    mux4to1 MUX (
```

```
        .F(clk),  
        .F2(F2),  
        .F4(F4),  
        .F8(F8),  
        .sel(sel),
```

```
        .clkOut(selected_clk)
```

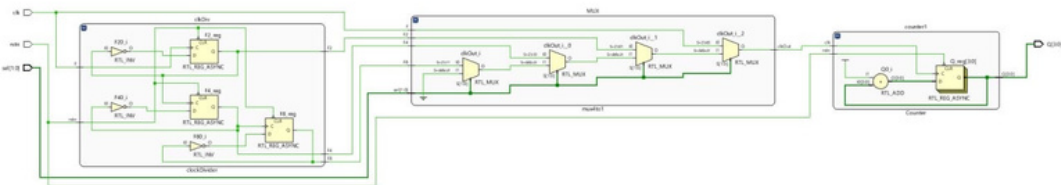
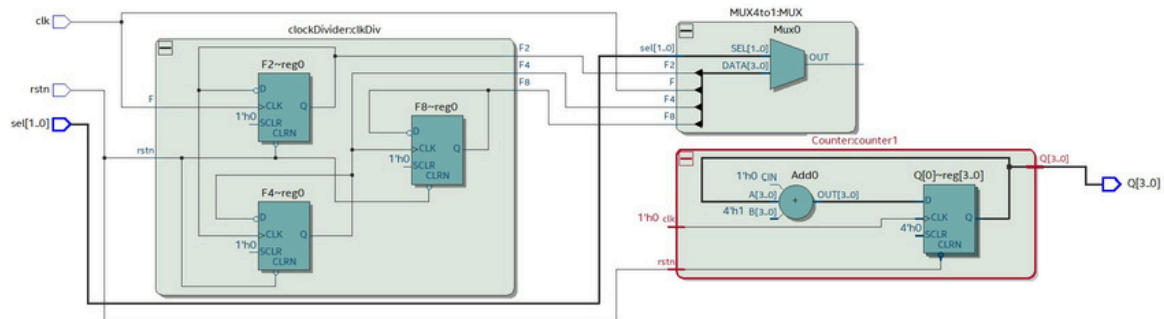
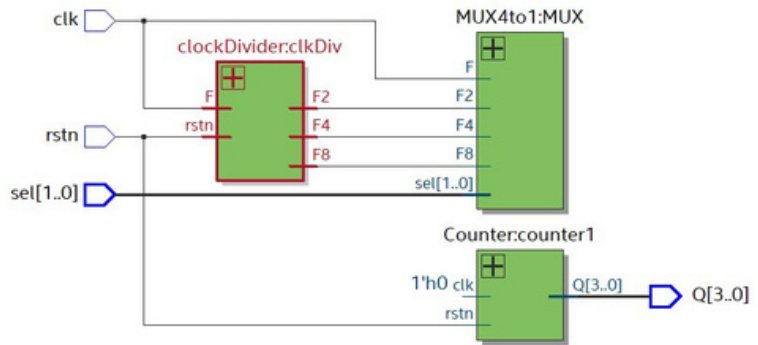
```
);
```

```
    Counter #(N) counter1(
```

```
        .clk(selected_clk),  
        .rstn(rstn),  
        .Q(Q)
```

```
);
```

```
endmodule
```



Task #11.1

```
`timescale 1ns/1ps
module top_tb;
```

```
localparam N = 4;
```

```
logic clk;
logic rstn;
logic [1:0] sel;
logic [N-1:0] Q;
```

```
top #(N) DUT (
    .clk(clk),
    .rstn(rstn),
    .sel(sel),
    .Q(Q)
);
```

```
initial clk = 0;
always #5 clk = ~clk;
```

```
initial begin
    rstn = 0;
    sel = 2'b00;
    #20;
    rstn = 1;
```

```
    sel = 2'b00;
    #200;
```

```
    sel = 2'b01;
    #200;
```

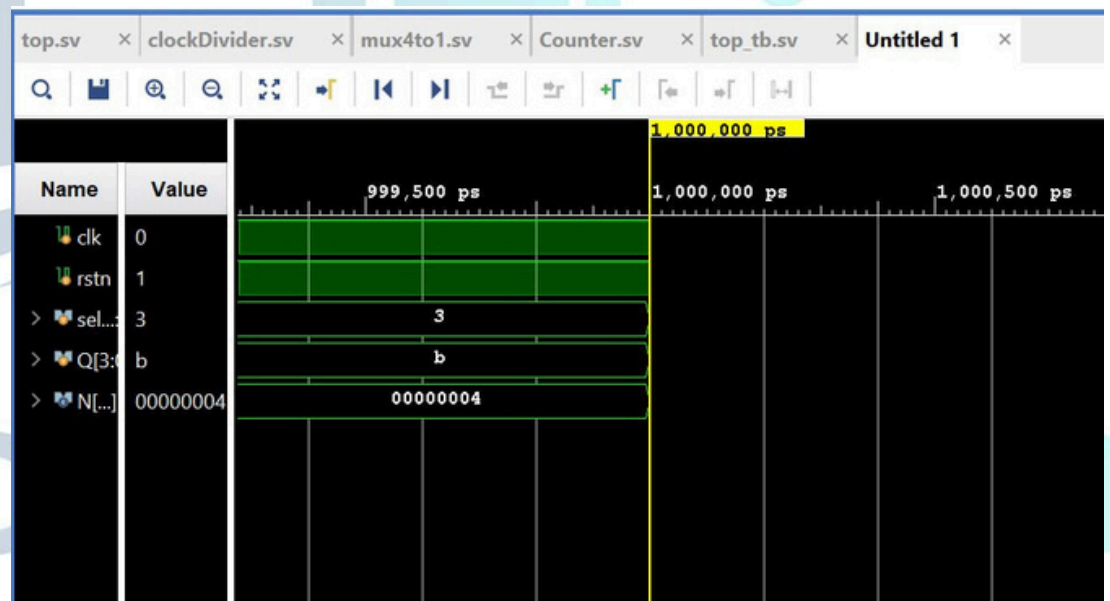
```
    sel = 2'b10;
    #400;
```

```
    sel = 2'b11;
    #800;
```

```
    $finish;
end
```

```
initial begin
    $monitor("Time=%0t sel=%b Q=%d",
    $time, sel, Q);
end
```

```
endmodule
```



Work distribution

Section	Almas	Atheer	Mjd	Maha	Jood
Report	✓	✓	✓	✓	✓
Task9.1	✓	✓	✓	✓	✓
Task9.2	✓	✓	✓	✓	✓
Task9.3	✓	✓	✓	✓	✓

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